

# Evaluating Evidence and Code Documentation in the Bank Customer Churn Prediction Case Study

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## Section One: Conclusions and Supporting Evidence

The purpose of a data science project is not simply to generate statistics and visualizations but to produce conclusions that are supported by evidence. When evaluating the Bank Customer Churn Prediction case study (Malit, n.d.), most of the narrative conclusions were supported by exploratory data analysis, visualizations, and summary statistics. However, several conclusions were stronger than the evidence presented.

One example involved the discussion of customer characteristics associated with churn. The notebook identified differences between customers who remained with the bank and customers who left. Visualizations demonstrated relationships between variables such as age, account balance, geography, and tenure. While these plots suggested patterns, they did not establish that those variables directly caused customers to leave the bank. The narrative occasionally implied stronger relationships than the evidence justified. Data visualizations can demonstrate associations, but they cannot independently establish causation. Additional analysis, controlled experimentation, or more advanced statistical testing would be required before concluding that a specific factor caused customer churn.

Another example involved geographic differences in churn behavior. The analysis indicated that customers from some geographic regions appeared more likely to leave the bank than others. While the charts showed variation among groups, the notebook did not provide statistical significance testing to determine whether the observed differences were meaningful or simply due to random variation within the sample. Without supporting statistical evidence, the conclusion should be interpreted as an observed pattern rather than a confirmed relationship.

This does not mean the analysis was incorrect. Rather, it highlights an important principle of data science: conclusions should be proportional to the strength of the evidence. Visualizations and descriptive statistics are useful for identifying potential relationships, but additional validation is often required before making stronger claims.

## Section Two: The Lack of Comments in the Programming Code

The notebook contains very few comments explaining what the programming code is doing. This is an inferior practice because code is written once but often read many times. Comments provide context, explain assumptions, and help future developers understand the purpose of the code. In collaborative environments, code without comments becomes difficult to maintain, debug, and extend.

Comments are particularly important in data science projects because the analysis often includes multiple stages such as cleaning, transforming, visualizing, and modeling data. A person

reviewing the notebook months later may understand Python syntax but still struggle to determine why a specific transformation or modeling choice was made.

One example from the notebook involved categorical encoding and model preparation. The code performed data transformations before training machine learning models. Although the code itself executed correctly, a reader unfamiliar with the specific functions would need to research each operation to understand the purpose of the transformation. The variable names and function calls indicated what the code accomplished, but they did not explain why the transformation was necessary.

A useful comment might have been:

- Convert categorical variables into numeric representations
- required for machine learning algorithms

Such a comment immediately explains the purpose of the code and reduces the effort required to understand the workflow.

Another example involved the creation of predictive models. The notebook trained classification algorithms to predict customer churn. While the code showed how the models were created, comments describing the reason for selecting a specific algorithm or explaining important parameter choices would improve readability and maintainability.

The absence of comments is especially problematic in notebook-based projects because notebooks frequently combine code, analysis, and narrative. Without adequate comments, future users may understand the results but struggle to reproduce or modify the analysis.

## Conclusion

The Bank Customer Churn Prediction case study provides a useful example of exploratory data analysis and predictive modeling. Most conclusions presented in the notebook were supported by visualizations and summary statistics. However, some interpretations were stronger than the available evidence justified because the analysis relied primarily on descriptive methods rather than statistical validation. Additionally, the lack of comments throughout the code reduced readability and maintainability. Effective data science requires both evidence-based conclusions and well-documented programming practices. Together, these principles improve transparency, reproducibility, and the overall quality of analytical work.

## References

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